



Preparing for Engineering Maths

Formulae Booklet

Differentiation

y	$\frac{dy}{dx}$
k (any constant)	0
ax	a
ax^n	$n \times ax^{n-1}$
$a e^{bx}$	$b \times a e^{bx}$
$a \ln(bx)$	$\frac{a}{x}$
$a \sin(bx)$	$b \times a \cos(bx)$
$a \cos(bx)$	$-b \times a \sin(bx)$
$a \tan(bx)$	$b \times a \sec^2(bx)$
$a \sinh(bx)$	$b \times a \cosh(bx)$
$a \cosh(bx)$	$b \times a \sinh(bx)$

Integration

y	$\int y \, dx$
k (any constant)	$kx + C$
ax^n	$\frac{ax^{n+1}}{n+1} + C \quad (n \neq -1)$
$a e^{bx}$	$\frac{a}{b} e^{bx} + C$
$\frac{a}{x}$	$a \ln(x) + C$
$a \sin(bx)$	$-\frac{a}{b} \cos(bx) + C$
$a \cos(bx)$	$\frac{a}{b} \sin(bx) + C$
$a \tan(bx)$	$\frac{a}{b} \ln \sec(bx) + C$
$a \sec^2(bx)$	$\frac{a}{b} \tan(bx) + C$
$a \sinh(bx)$	$\frac{a}{b} \cosh(bx) + C$
$a \cosh(bx)$	$\frac{a}{b} \sinh(bx) + C$

The Quadratic Formula

For the quadratic equation $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Rules of Indices

$$a^m \times a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^m \div a^n = a^{m-n}$$

$$\frac{A}{a^n} = Aa^{-n}$$

$$a^0 = 1$$

$$a^1 = a$$

$$\sqrt[n]{a} = a^{\frac{1}{n}}$$

$$\sqrt[n]{a^m} = a^{\frac{m}{n}}$$

$$(a^m)^n = a^{m \times n}$$

Rules for Fractions

Multiplying Fractions

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Dividing Fractions

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Triangle Geometry

Right angled triangles

Pythagoras' theorem:

$$a^2 = b^2 + c^2$$

where a is the hypotenuse and b, c are the other two sides.

General rules for triangles

Sine rule:

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

Cosine rule:

$$a^2 = b^2 + c^2 - 2bc \cos(A) \quad \text{or} \quad \cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$$